# 04INDTSTD1P3

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**Problem**

The game of *pebbles* is played on an infinite board of lattice points $(i, j)$. Initially there is a *pebble* at $(0, 0)$. A move consists of removing a *pebble* from point $(i, j)$and placing a *pebble* at each of the points $(i+1, j)$ and $(i, j+1)$ provided both are vacant. Show taht at any stage of the game there is a *pebble* at some lattice point $(a, b)$ with $0 \leq a + b \leq 3$.

**Solution.** Assign the pebble at $(0, 0)$ with a "mass" of 1. Whenever a pebble at $(i, j)$ is divided and split into $(i + 1, j)$ and $(i, j + 1)$, half of the pebble's mass is kept on each of the points.

Observe that the total mass of the pebbles is 1 and any pebble at $(i, j)$ has a mass of $\frac{1}{2^{i+j}}$.

If at any stage there are no pebbles on $(a, b)$ with $0 \leq a + b \leq 3$ then the total mass of pebbles on the infinite board can be at most

$$\left(\sum_{i=0}^{\infty} \frac{1}{2^i}\right)\left(\sum_{j=0}^{\infty} \frac{1}{2^j}\right) - 1 - \frac{2}{2} - \frac{3}{4} - \frac{4}{8} = \frac{3}{4}.$$

Since the total mass of the pebbles is 1, this is impossible! So at any stage of the game there is a *pebble* at some lattice point $(a, b)$ with $0 \leq a + b \leq 3$.